

From Physics to Selective Generation: Conditional Brownian Bridges for Sparse S–N Curve Reconstruction and Fatigue Decision-Making

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ABSTRACT

Dense S–N characterization consumes specimens and long-duration tests, yet early materials decisions are often made from only two to four failures. The sparse curves separate naturally into two regimes: a single-slope Basquin state is already informative for many cases, while knees, curvature and one-sided support create a difficult subset whose engineering quantities depend on the missing curve shape. We cast completion as a selective transformation problem and introduce a two-stage selective physics-anchored conditional Brownian bridge (S-PA-CBB). Stage I scores curve difficulty from sparse support and a provisional bridge posterior. Stage II retains the physical mean for physics-sufficient curves and applies a validation-shrunk residual transformation to the difficult subset, preserving posterior deviations and cross-stress covariance in both paths. Under identical sparse masks, S-PA-CBB achieves the lowest hidden-point error in all nine pooled database–budget settings across AM2022, CMA2022 and Weld2025. Target-excluded leave-one-database-out evaluation retains the advantage in eight of nine settings, with gains increasing as CMA2022 moves from two to three and four observations. Selective correction reduces raw-generator complete-grid error in every held-out database setting, while repeated samples improve proper distributional scores and stabilize covariance-guided experiment selection. Split-conformal calibration converts the generated spread into coverage-controlled intervals. The resulting framework links sparse physical fitting, selective residual generation and engineering decisions in one complete-curve representation.

1. Introduction

An S–N curve compresses a costly fatigue programme into a form that engineers can invert for fatigue strength, integrate for damage and interrogate when choosing the next specimen. The low-stress portion is especially expensive because failures may require millions of cycles. New alloy states, build orientations, heat treatments, surface conditions and load ratios multiply that burden. Basquin's log–log relation remains the natural first description of this curve [1], while modern reviews place it within a broader family of stress–life and probabilistic design models [2, 3]. Bayesian and probabilistic formulations quantify slope and scatter from limited tests [4, 5]. Sparse-data methods further improve parameter inference and design reliability [6, 7], and recent reviews summarize their role across fatigue regimes [8].

Materials screening creates a sharper version of this problem. Two to four failures may be available, yet the required outputs—fatigue strength at a design life, remaining life at a service stress, integrated damage and the next informative stress—depend on the entire curve. Machine-learning reviews document the growing use of composition, process and loading descriptors in fatigue prediction [9, 10]. Representative models address general life regression and cracked components [11, 12], mean-stress and thermomechanical effects [13, 14], and defect-aware or physics-constrained learning

[15, 16]. Alloy-specific studies extend this progress to very-high-cycle fatigue and quantify predictive gains over physical baselines [17, 18]. These approaches are effective point predictors; sparse curve completion adds the requirement that all stresses share one coherent conditional trajectory.

The key observation is that sparse completion has two physical regimes. Many curves remain close to a monotone single-slope relation, so the sparse Basquin fit already carries most of the useful structure. Curves with shoulders, knees, local curvature, regime-dependent slopes or one-sided observations contain a residual shape that moves engineering quantities. A flexible correction is valuable in this second regime and unnecessary in the first. Physics-informed learning provides the general principle of preserving known structure while estimating its discrepancy [19]. Physics-informed neural networks formalize this coupling between governing relations and learned fields [20], while materials machine learning shows the value of physically meaningful intermediate representations [21]. We therefore organize sparse completion as a decision: assess the physical state, then transform the subset whose residual geometry carries additional information. Figure 1 summarizes this mechanism.

Conditional generation supplies the distribution required for that transformation. Denoising diffusion models learn complex conditional distributions through progressive corruption and reconstruction [22]. Surveys describe their model families and practical design choices [23, 24], while recent theory identifies the approximation errors that accumulate along the reverse trajectory [25, 26]. Time-series diffusion extends these ideas to ordered one-dimensional signals [27, 28], and fast solvers reduce the number of network

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evaluations [29]. Brownian-bridge diffusion is particularly suitable here because it connects two meaningful endpoints [30]; constrained inpainting shows how measured entries can remain fixed throughout sampling [31]. In our construction, the terminal endpoint is the sparse Basquin state, a one-dimensional U-Net resolves curve geometry [32], and a zero-initialized ControlNet path injects anchors and fatigue conditions [33].

The scientific value of generation lies in the distribution of complete trajectories. Proper scoring rules assess predictive distributions [34]; interval scores measure calibration and sharpness [35], and variogram scores probe dependence across stress locations [36]. Split-conformal prediction then maps validation residuals to coverage-controlled intervals [37]. The resulting ensemble can support sequential experiments through its cross-stress covariance. Active-learning studies in materials already show how uncertainty can guide targeted measurement and discovery [38, 39]. Recent autonomous experimentation frameworks extend this principle to closed-loop materials campaigns [40], while critical evaluations emphasize matching the acquisition function to the scientific objective [41].

Three public resources provide distinct domains for testing this complete-curve view. AM2022 covers additively manufactured metals [42]; CMA2022 contains metallic glasses and multi-principal-element alloys [43]; Weld2025 contributes welded joints with geometry and treatment metadata [44]. We harmonize stress definitions and curve grouping, audit article- and curve-level duplicates, preserve database identity and evaluate both pooled and target-excluded transfer. This protocol connects the proposed mechanism to three questions: whether the selective correction improves hidden measurements, whether generated trajectories form a useful distribution, and whether that distribution changes an engineering decision.

The study follows one central question: *when should a sparse physical S-N state be retained, and when does a generated residual add engineering information?* Four contributions answer it:

- A two-regime formulation separates physics-sufficient curves from curves requiring a structured residual transformation.
- A distributional audit combines proper scores, noise intervention, sample-count sensitivity, conformal calibration and cross-stress covariance.
- A mask-aligned three-database benchmark, true LODO evaluation and five downstream decisions connect reconstruction to fatigue engineering use.
- The resulting S-PA-CBB architecture joins a Basquin intermediate state, validation-fitted difficulty screening and a ControlNet-conditioned Brownian bridge that preserves posterior covariance along both decision paths.

Section 2 specifies the model and inference algorithm. Section 3 defines the data protocol, compact reconstruction

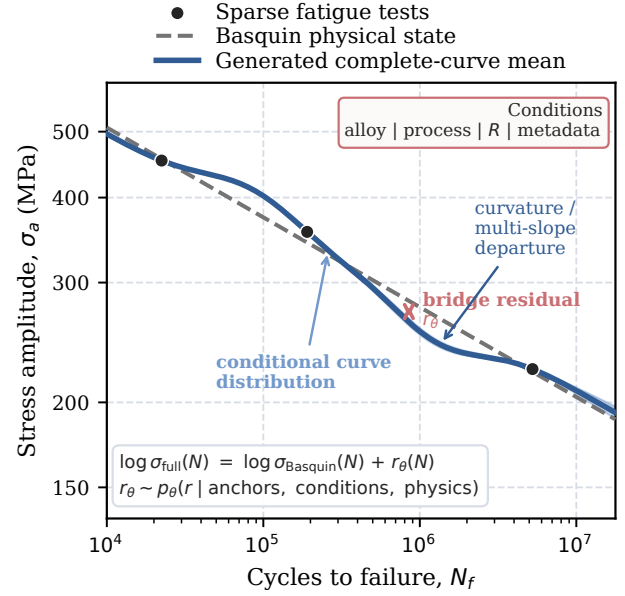


Figure 1: Conceptual mechanism of two-stage selective transformation. Sparse tests define a Basquin state and a difficulty assessment. Physics-consistent curves retain that state; flagged difficult curves receive a generated residual transformation and a distribution of complete S-N curves.

benchmark and component check. Section 4 reports the validation-selected engineering tasks and active experiment selection.

2. Method

2.1. Problem definition and notation

For fatigue curve c , let s denote a scalar stress coordinate after conversion to a common MPa convention, $x = \log_{10} s$, and $y = \log_{10} N$ the log cycles to failure. The complete curve is represented on a monotonically increasing grid $\mathbf{x}_c = (x_{c1}, \dots, x_{cG})$ with $G = 32$. Grid limits are curve-specific during preprocessing and are mapped to $[-1, 1]$ for the network. The target vector is $\mathbf{y}_{0,c} \in \mathbb{R}^G$.

At test time, only $k \in \{2, 3, 4\}$ experimental failures are revealed. Their values are projected to the common grid, producing a sparse signal $\mathbf{o}_c$, a mask $\mathbf{m}_c \in \{0, 1\}^G$ and the original off-grid coordinates used for exact evaluation. Curve-level metadata are encoded as $\mathbf{q}_c$. Depending on the database, $\mathbf{q}_c$ can contain alloy family, nominal composition, manufacturing route, heat treatment, surface treatment, load ratio, frequency, temperature, specimen class, weld category and stress-definition flags. Missing categorical values receive an explicit unknown token; continuous variables are accompanied by missingness indicators.

The desired output is the conditional distribution

$$p_{\theta}(\mathbf{y}_0 | \mathbf{o}, \mathbf{m}, \mathbf{x}, \mathbf{q}, \mathbf{y}_{\text{phys}}), \quad (1)$$

whose draws pass through the revealed observations after projection, remain compatible with the material/test condition

Workflow figure placeholder

Three databases → harmonization and duplicate audit → strict splits
 → sparse anchors and metadata → Basquin physical state
 → controlled Brownian bridge → complete-curve ensemble
 → conformal calibration → benchmark and downstream decisions

Figure 2: Placeholder for the full method workflow. The final vector figure will distinguish offline database preparation, model training and inference for a new sparse curve. It will also show that external databases remain unavailable to the model in zero-shot evaluation.

and represent plausible full-curve shapes. The predictive mean, quantiles, fatigue strengths and acquisition scores are derived from the same ensemble.

2.2. Overview and large workflow figure

Figure 2 is reserved for the final method diagram. The intended left-to-right flow is: three reported public databases; unit and definition harmonization; DOI/title/curve duplicate audit; source-disjoint split construction; sparse-mask simulation; Basquin physical-state estimation; ControlNet-conditioned Brownian bridge; ensemble of full S–N curves; conformal calibration; point, curve, distributional and downstream evaluation. The final artwork should visually separate information available during training from information available for a new sparse curve.

2.3. Data harmonization and curve construction

The bridge requires comparable one-dimensional curves, so raw records pass through a definition-aware conversion layer. Stress amplitude is retained directly. Stress range is divided by two when the database explicitly reports a range and the load convention permits the conversion. Maximum stress is converted using the reported load ratio $R = \sigma_{\min}/\sigma_{\max}$, giving $\sigma_a = (1 - R)\sigma_{\max}/2$. Ambiguous stress definitions are assigned to a separate stress-type domain or omitted from the common-amplitude benchmark. Reversals are converted to cycles when the original definition is explicit.

The completion likelihood uses failure points, while runouts remain marked as censored observations for consistency checks and future likelihood extensions. Within one database, points form a curve when material state, processing, specimen, environment, stress ratio, frequency and temperature agree at the available resolution. Curves require at least six failures, four unique stresses, log-stress span of at least 0.04 and a fitted negative slope in $[-25, -0.05]$. Replicates at the same stress are preserved for raw-data auditing; a robust centre constructs the 32-point reference representation.

Cross-database duplicates are detected hierarchically. Exact DOI matches are checked first, followed by normalized title and author/year matches. Candidate curves within

matched articles are then compared by alloy, R , temperature, process labels and rounded (S, N) pairs. Any matched record is assigned to one source group before splitting and is never counted as an independent test curve in another database.

2.4. Basquin physical state

The physical state is fitted exclusively from the revealed anchors. The Basquin relation can be written

$$y = a + bx, \quad b < 0. \quad (2)$$

For $k \geq 3$, (a, b) are estimated by robust regression. For $k = 2$, the two points identify a line but the slope can be unstable when their stresses are close. We therefore shrink the sparse slope toward a training-only conditional prior:

$$\hat{b} = \lambda_c b_{\text{sparse}} + (1 - \lambda_c) b_{\text{prior}}(q), \quad \lambda_c = \frac{\Delta x_c^2}{\Delta x_c^2 + \tau_b^2}, \quad (3)$$

where Δx_c is the observed log-stress span and τ_b is selected on the validation set. The intercept is then chosen to minimize robust anchor residuals. Slope clipping enforces negative, numerically stable extrapolation. Evaluation on the grid produces $\mathbf{y}_{\text{phys}}$, the physical intermediate representation evaluated through component ablation.

The physical state serves three roles. First, it supplies a monotone low-frequency trend. Second, it provides an endpoint for the Brownian bridge, making the stochastic path conditional by construction. Third, it converts a hard absolute-learning problem into residual transport. The model must learn $\mathbf{y}_0 - \mathbf{y}_{\text{phys}}$, which is smaller and more comparable across strength scales than $\mathbf{y}_0$ itself.

2.5. Brownian-bridge forward process

Let $t \in (0, 1)$ and $m_t = t$. A forward bridge between the complete curve and the physical endpoint is

$$\mathbf{y}_t = (1 - m_t)\mathbf{y}_0 + m_t\mathbf{y}_{\text{phys}} + \sigma \sqrt{m_t(1 - m_t)}\boldsymbol{\epsilon}, \quad \boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}). \quad (4)$$

The variance is zero at both endpoints and largest near the middle of the path. This differs from a conventional

unconditional diffusion model, where the terminal state is approximately Gaussian noise. Here the terminal state is a physically interpretable curve, and reverse sampling transports that curve toward the conditional data distribution.

The network returns three length- G heads: predicted noise $\hat{\epsilon}$, a direct clean-curve estimate $\hat{\mathbf{y}}_0^{\text{dir}}$, and log variance $\hat{\nu}$. A score-derived clean estimate follows from rearranging Eq. (4). The direct and score-derived estimates are blended as a function of time: the direct head stabilizes early reverse steps near the physical endpoint, while the score-derived estimate receives more weight near the data endpoint.

2.6. One-dimensional U-Net and control pathway

The base network is a one-dimensional U-Net with base width 48 and three spatial resolutions. Each resolution contains residual convolutional blocks, group normalization and a time embedding. Downsampling expands the receptive field so that local anchor information can affect distant stresses; skip connections preserve location-specific detail. The final heads operate at the original 32-point resolution.

The control encoder mirrors the backbone encoder. Its channels concatenate the sparse grid values, observation mask, normalized coordinate, physical state and broadcast metadata embedding. Following the ControlNet principle [33], control residuals enter the frozen/pretrained backbone through initially zero-valued projections. The conditional branch therefore begins as an identity perturbation of the base transport process. In the second training stage, the control parameters and selected backbone layers are optimized jointly.

Metadata are encoded by type. Categorical fields use learned embeddings with an unknown category. Continuous fields are standardized using training statistics and paired with presence bits. Alloy composition, where available, is mapped to a fixed element vector; otherwise an alloy-family embedding is used. The principal zero-shot setting uses transferable material and test descriptors, while database identity is reserved for diagnostic domain-conditioning experiments.

At every reverse step, revealed observations are projected back into the current state. For an on-grid anchor, the replacement is exact. For an off-grid anchor, the two adjacent grid values are corrected by the minimum-norm linear update that makes their interpolation equal to the observation. This projection enforces observation fidelity as an algorithmic constraint.

2.7. Training objective

The loss combines four terms:

$$\mathcal{L} = \mathcal{L}_\epsilon + \lambda_0 \mathcal{L}_0 + \lambda_\Delta \mathcal{L}_\Delta + \lambda_c \mathcal{L}_{\text{cons}}, \quad (5)$$

$$\mathcal{L}_\epsilon = \frac{1}{G} \sum_j [\exp(-\hat{\nu}_j)(\epsilon_j - \hat{\epsilon}_j)^2 + \hat{\nu}_j], \quad (6)$$

$$\mathcal{L}_0 = \|\hat{\mathbf{y}}_0^{\text{dir}} - \mathbf{y}_0\|_2^2, \quad (7)$$

$$\mathcal{L}_\Delta = \|\Delta \hat{\mathbf{y}}_0^{\text{dir}} - \Delta \mathbf{y}_0\|_1, \quad (8)$$

$$\mathcal{L}_{\text{cons}} = \text{SmoothL1}(\hat{\mathbf{y}}_0^{\text{dir}}, \hat{\mathbf{y}}_0^{\text{score}}). \quad (9)$$

The heteroscedastic noise term learns location-dependent stochastic scale. The clean loss stabilizes the small-grid problem, and the first-difference loss discourages spurious oscillations while retaining multi-slope and plateau behaviour. A hard monotone projection is evaluated separately as a decision-time variant.

Training uses two stages. The base bridge is trained for 60 epochs. The control branch is then attached and trained for 80 epochs. AdamW uses learning rate 2×10^{-3} , weight decay 10^{-4} , batch size 48 and gradient-norm clipping at 2.0. Sixteen sparse masks are sampled per training curve across $k = 2, 3, 4$. Seeds fix the dataset split, mask stream and parameter initialization. Distributed training uses seven Ascend devices; synchronized per-curve inference latency is reported separately in the sampler ablation.

2.8. Reverse sampling and posterior ensemble

All main, external, leave-database-out and downstream results use one frozen sampler: three reverse steps, 32 independent trajectories and transition-noise scale $\eta = 0.5$. The sampler begins exactly at the physical endpoint, whose Brownian-bridge variance is zero, predicts the clean state at each step, advances toward $t = 0$, injects Gaussian stochasticity at non-terminal intermediate states and re-imposes the observed anchors. Split-conformal calibration is fitted on validation curves after fixing this sampler. Longer-step and ensemble-size sweeps serve as sensitivity analyses around the common headline protocol. Algorithm 1 specifies inference for one sparse curve.

2.9. Stage I difficulty screening and Stage II residual transformation

Unconditional residual generation can over-correct a curve whose sparse physical state is already adequate. The deployed model therefore treats completion as a two-stage decision. In Stage I, a provisional physics-anchored bridge posterior assesses difficulty. Prospective quantities available at inference—generated tail-residual magnitude, observed stress-span coverage and posterior spread—form a scalar score. A validation-fitted threshold τ_k maps that score to $g \in \{0, 1\}$: $g = 0$ denotes a physics-sufficient curve and $g = 1$ denotes a curve requiring transformation. Gate fitting uses validation curves and prospective quantities exclusively.

Stage II either retains or transforms the physical state. Let $\{\mathbf{y}_{\text{raw}}^{(r)}\}_{r=1}^M$ be the provisional ensemble, $\bar{\mathbf{r}} = M^{-1} \sum_r \mathbf{y}_{\text{raw}}^{(r)} - \mathbf{y}_{\text{phys}}$ its mean residual, and $\mathbf{d}^{(r)} = \mathbf{y}_{\text{raw}}^{(r)} - \bar{\mathbf{r}}$ its zero-mean posterior deviations. For each sparse budget, the transformation strength α_k is also fitted on validation curves. The final posterior is

$$\mathbf{y}_{\text{sel}}^{(r)} = \mathbf{y}_{\text{phys}} + g \alpha_k \bar{\mathbf{r}} + \mathbf{d}^{(r)}. \quad (10)$$

Thus Stage I decides whether transformation is warranted, while Stage II applies a validation-chosen fraction of the residual and preserves stochastic deviations and cross-stress covariance. When $g = 0$, the mean reverts to the physical state while uncertainty remains available for calibration and

Algorithm 1 Physics-anchored conditional generation for one sparse S–N curve

Require: Observations $\mathcal{O} = \{(s_i, N_i)\}_{i=1}^k$; metadata $\mathbf{q}$; trained model f_θ ; grid size G ; reverse steps T ; ensemble size M

Ensure: Posterior samples $\{\mathbf{y}_0^{(r)}\}_{r=1}^M$ and calibrated intervals

- 1: Convert explicit stress/life definitions to $x_i = \log_{10} s_i$ and $y_i = \log_{10} N_i$
- 2: Construct grid $\mathbf{x}$, sparse signal $\mathbf{o}$ and observation mask $\mathbf{m}$
- 3: Estimate $(\hat{a}, \hat{b})$ with Eq. (3) and evaluate $\mathbf{y}_{\text{phys}} = \hat{a} + \hat{b}\mathbf{x}$
- 4: Encode control tensor $\mathbf{h} = \text{Encode}(\mathbf{o}, \mathbf{m}, \mathbf{x}, \mathbf{q}, \mathbf{y}_{\text{phys}})$
- 5: **for** $r = 1$ to M **do**
- 6: Initialize $\mathbf{z}_T \leftarrow \mathbf{y}_{\text{phys}}$ exactly at the physical endpoint
- 7: **for** $j = T, T-1, \dots, 1$ **do**
- 8: $(\hat{\epsilon}, \hat{\mathbf{y}}_0^{\text{dir}}, \hat{\mathbf{v}}) \leftarrow f_\theta(\mathbf{z}_j, \mathbf{h}, t_j)$
- 9: Compute score-derived $\hat{\mathbf{y}}_0^{\text{score}}$ and blend the two clean estimates
- 10: Advance $\mathbf{z}_{j-1}$ with the Brownian-bridge reverse update and predicted variance
- 11: Project the revealed observations back onto $\mathbf{z}_{j-1}$
- 12: **end for**
- 13: Store $\mathbf{y}_0^{(r)} \leftarrow \mathbf{z}_0$
- 14: **end for**
- 15: Compute ensemble mean, raw quantiles and curve-derived downstream quantities
- 16: Apply validation-fitted conformal residual quantiles while preserving the generated mean
- 17: **return** posterior ensemble and calibrated prediction bands

acquisition. When $g = 1$, the difficult curve receives a structured residual transformation. The unconditional PA-CBB is retained as the ablation obtained by transforming every curve.

2.10. Split-conformal calibration

Raw ensemble quantiles are diagnosed before correction. For each target coverage $1 - \alpha$, the validation set supplies nonconformity scores based on the distance of a held-out observation from the raw central interval. The finite-sample conformal quantile expands the lower and upper bounds symmetrically in $\log_{10} N$ space. Calibration parameters are fitted separately by sparsity level, and subgroup calibration is reported when its validation sample supports estimation. Expansion widths are fixed from validation data before test evaluation.

This separation assigns distinct roles to generation and calibration. The stochastic head supplies structured spread, while the conformal layer aligns empirical coverage with the requested level. Raw coverage, conformal coverage, interval width and interval score are reported together to quantify both calibration and sharpness.

3. Experimental design and results

3.1. Three-database benchmark

Table 1 defines the role of every database. A common conversion and filtering pipeline maps the different release-level objects to eligible curve counts after local download and audit.

For AM2022, the present filter retains curves with at least six non-runout failures, at least four distinct stresses, sufficient log-stress span and a negative fitted slope. Grouping by source paper yields 609 training, 203 validation and 203 test curves. No paper contributes to more than one split. The

same test curve receives a fixed mask for every method, and metrics are computed on exactly the same hidden points.

External evaluation has two levels. The pooled benchmark estimates database-resolved performance under one harmonized training protocol. In *leave-database-out* (LODO) evaluation, the target database is excluded from both training and validation; the held-out database is opened only after model, gate and correction-strength selection. LODO is reported for AM2022, CMA2022 and Weld2025 because each supports a database-level estimate.

3.2. Data coverage and uncertainty structure

The harmonized resources occupy different stress–life regions, while AM2022 and Weld2025 dominate the eligible-curve count. This imbalance makes macro-averaged, database-resolved reporting necessary: a pooled point average would otherwise be governed by the welded-joint resource.

Figure 3 summarizes the public-data condition space before model comparison. The parallel coordinates show curve support and test conditions, while the lower panels expose alloy, process and publication-year imbalance. These descriptors motivate source-aware splitting and prevent a large database from being mistaken for broad physical coverage.

The benchmark treats database shift as a first-class variable. AM2022 and CMA2022 use related literature-derived schemas but cover different material families, whereas Weld2025 changes both the physical object and metadata hierarchy. This separation allows an in-protocol pooled result to be distinguished from target-excluded transfer.

Figure 4 restores the strict-split uncertainty audit used to examine whether generated spread is informative. It separates point accuracy, raw interval coverage, conformal coverage and interval width, and it tests whether wider generated intervals coincide with larger errors. The figure is

Table 1

Public fatigue resources after common curve construction. Release statistics are abbreviated; eligible curves are computed by the present preprocessing pipeline.

Database	Scope	Eligible	Use
FatigueData-AM2022 [42]	Additively manufactured metals with process and test metadata	1,015	Development split and alloy-held-out tests
FatigueData-CMA2022 [43]	Metallic glasses and multi-principal-element alloys	105	Cross-material and pooled evaluation
Welded-joint database [44]	Weld, geometry and post-treatment conditions	2,817	Large domain-shift evaluation

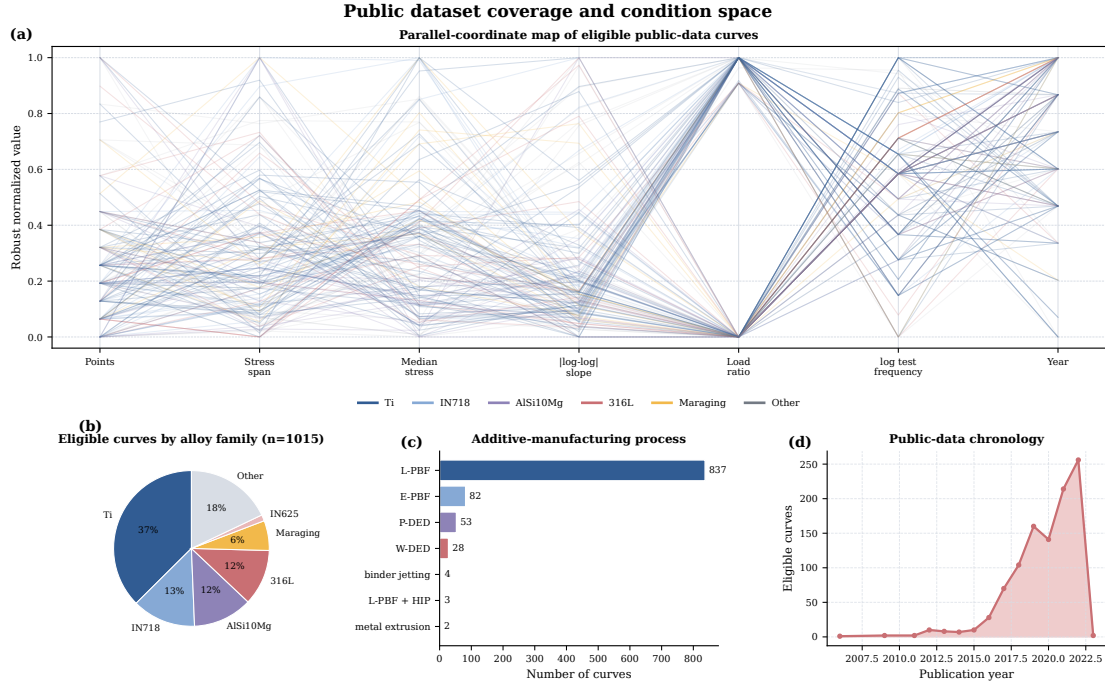


Figure 3: Question: What condition space is represented by the public fatigue data? The parallel-coordinate map summarizes curve and test descriptors; the lower panels show alloy-family composition, manufacturing routes and publication chronology. **Finding:** coverage is broad but strongly imbalanced, with a few alloys and processes dominating the eligible curves.

a calibration diagnostic across the source-paper holdout and leave-one-alloy-out partitions; the frozen three-step headline values are reported separately in Section 3.6.

3.3. Compared methods

The main comparison spans three baseline families. Linear, polynomial and PCHIP interpolation test whether simple curve geometry is sufficient. k NN, random forest and ExtraTrees test conventional data-driven regression without a physical intermediate state. Direct U-Net tests deterministic deep completion with the same grid and input information. All methods receive the same held-out curves and hash-derived masks. Basquin-only and a matched bridge without the physical state remain component ablations because Basquin supplies the physical endpoint of the proposed method.

Hyperparameters for learned baselines are fixed before opening the test curves, using validation performance and the pre-declared settings released with the evaluation code. Direct U-Net uses the same grid representation and comparable channel width as the proposed backbone. Matched architecture

and physics ablations use shared splits, masks and seeds; between-seed standard deviations and paired curve-bootstrap confidence intervals quantify their uncertainty.

3.4. Evaluation metrics

Hidden-point performance uses MAE, median absolute error and factor-of-two accuracy. Factor-of-two accuracy is the fraction with $|\hat{y} - y| \leq \log_{10} 2$ and has a direct multiplicative-life interpretation. Full-grid evaluation uses mean absolute error against the reference interpolant, global log-log slope error and monotonicity-violation rate.

Probability quality uses CRPS for individual hidden points, WIS for central intervals, and Energy Score for the entire 32-dimensional curve. Variogram Score emphasizes dependence between stress locations. Calibration is reported as empirical coverage at 50%, 80%, 90% and 95%, mean width, absolute coverage error and simultaneous curve coverage. Average pairwise L_1 distance summarizes diversity, while proper scores determine distributional quality.

Statistical comparisons are paired by curve and mask. For each metric, 10,000 group-bootstrap replicates resample

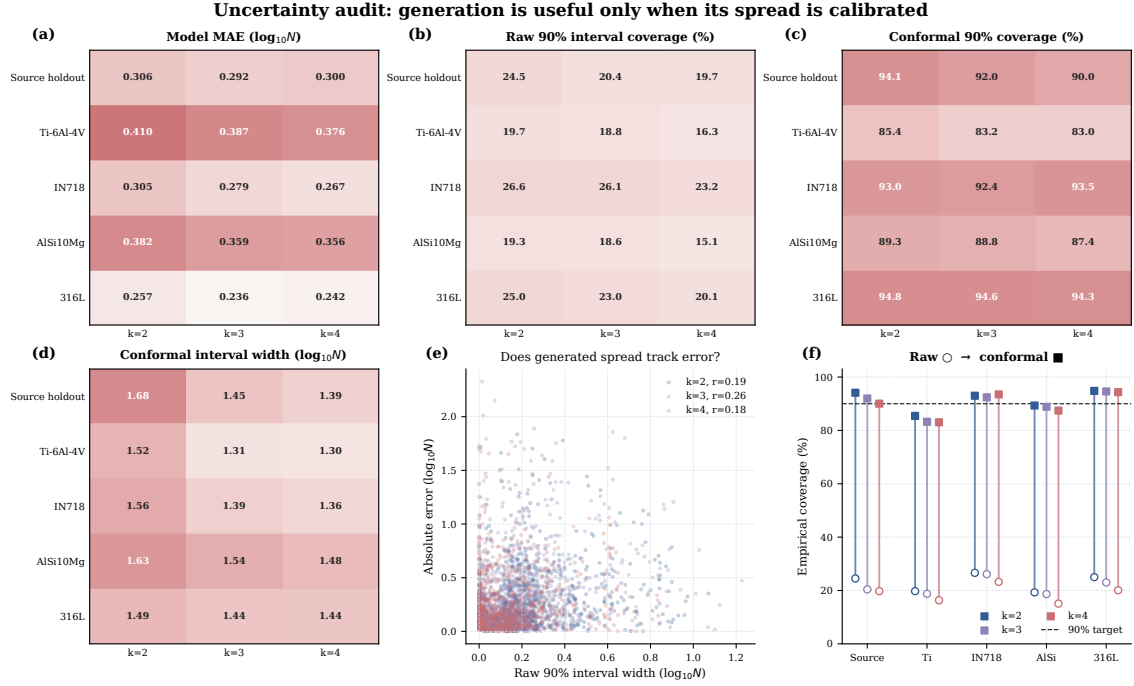


Figure 4: Question: Does stochastic spread behave as calibrated predictive uncertainty across strict splits? Panels (a–d) compare point error, raw coverage, conformal coverage and calibrated width; panel (e) relates raw interval width to absolute error; panel (f) shows the change from raw to conformal coverage. **Finding: generated spread contains a weak error signal but is substantially under-dispersed before calibration. Split-conformal expansion restores coverage toward its target, with residual shortfall under the strongest alloy shift.**

source papers. Improvements are assigned when the corresponding 95% confidence interval excludes zero, and multiple comparisons across baselines use Holm correction. Database-resolved reporting gives AM2022, CMA2022 and Weld2025 independent visibility despite their different sizes.

3.5. Expanded reconstruction benchmark

The reported pooled test contains 789 untouched curves after validation-only gate fitting: 188 AM2022, 20 CMA2022 and 581 welded-joint curves. Table 2 gives the aligned AM2022 comparison under six non-physics baselines. S-PA-CBB has the lowest full-grid error at all three sparse budgets. It leads seven of eight hidden-point metrics with two observations, six of eight with three observations and all eight with four observations; random forest or ExtraTrees leads the remaining RMSE and upper-tail-error cells.

Figure 5 summarizes the promoted main model. It reports the validation gate, pooled complete-curve error against both the physical state and unconditional bridge, activation frequency and raw-versus-conformal calibration. Unlike a comparison based only on hidden points, it directly tests the quantity generated by the model: the complete curve.

3.6. Distributional evaluation and calibration

For $k = 2, 3, 4$, pooled CRPS is 0.292, 0.264 and 0.245, while Energy Score is 0.277, 0.236 and 0.213 under the frozen three-step sampler and selective mean translation. Energy Score is especially relevant because a sampled curve is evaluated as one 32-dimensional object.

The raw ensemble is sharply under-dispersed. Nominal 90% intervals cover 5.6%, 4.5% and 4.4% of hidden points. Validation-only split-conformal expansion raises coverage to 87.9%, 87.2% and 88.8%, with mean interval widths of 1.302, 1.208 and 1.170 log cycles. These results identify the generated ensemble as the structured uncertainty carrier and conformal expansion as the coverage-control layer.

Figure 6 provides a qualitative success-case view across two, three and four revealed tests and the three principal databases. Thin trajectories and the 90% band expose the generated distribution, while the reference, posterior mean and observed anchors locate its support. Cases are selected by a deterministic rule requiring low final grid error and, for activated cases, a positive correction gain; Table 2 supplies the corresponding aggregate rather than case-selected evidence.

The gallery also exposes the action of the selective layer. Each sparse-budget row contains three gate-active examples whose blue mean departs from Basquin after validation evidence supports a residual correction, followed by one case in which the accurate sparse physical mean is retained while posterior trajectories are preserved. Both paths therefore produce decision-ready uncertainty, with the gate controlling the central transformation.

Across the gallery, observed anchors are re-imposed exactly after every reverse step. Differences between the selective mean and the reference therefore occur between or beyond measured stresses. The deliberately selected success cases illustrate attainable reconstruction quality and are not

Table 2

AM2022 source-paper-held-out comparison under identical sparse masks. The main comparison contains non-physics baselines; component variants are reported in the ablation study. Bold and underline mark first and second place.

Method	MAE ↓	RMSE ↓	MedAE ↓	P90AE ↓	R^2 ↑	Pearson r ↑	F2 ↑	F3 ↑
$k = 2$ observed tests								
kNN	0.465	0.639	0.342	1.026	0.622	0.802	0.458	0.630
RF	<u>0.384</u>	0.531	<u>0.275</u>	<u>0.867</u>	<u>0.738</u>	<u>0.860</u>	<u>0.536</u>	<u>0.719</u>
ET	<u>0.391</u>	0.537	<u>0.275</u>	<u>0.908</u>	<u>0.732</u>	<u>0.857</u>	<u>0.526</u>	<u>0.699</u>
Linear	0.852	1.518	<u>0.346</u>	2.626	-1.137	0.454	0.458	0.594
Poly.	0.845	1.507	0.346	2.599	-1.105	0.457	0.458	0.595
PCHIP	0.852	1.518	0.346	2.626	-1.137	0.454	0.458	0.594
S-PA-CBB (ours)	0.373	<u>0.535</u>	0.247	0.866	0.761	0.873	0.558	0.726
$k = 3$ observed tests								
kNN	0.441	0.595	0.326	0.973	0.672	0.837	0.472	0.656
RF	<u>0.364</u>	<u>0.512</u>	<u>0.247</u>	0.830	0.757	0.871	<u>0.562</u>	<u>0.727</u>
ET	<u>0.370</u>	0.508	<u>0.267</u>	<u>0.847</u>	<u>0.761</u>	<u>0.875</u>	<u>0.543</u>	<u>0.726</u>
Linear	0.705	1.299	0.286	2.104	-0.564	0.559	0.518	0.654
Poly.	0.914	1.608	0.352	2.860	-1.396	0.433	0.456	0.580
PCHIP	1.005	1.749	0.360	3.463	-1.835	0.401	0.445	0.571
S-PA-CBB (ours)	0.360	0.516	0.241	0.885	0.777	0.882	0.582	0.741
$k = 4$ observed tests								
kNN	0.424	0.566	0.319	0.928	0.710	0.853	0.479	0.672
RF	<u>0.374</u>	0.529	<u>0.255</u>	0.870	0.746	0.865	<u>0.561</u>	<u>0.728</u>
ET	<u>0.382</u>	<u>0.523</u>	<u>0.277</u>	<u>0.865</u>	<u>0.752</u>	<u>0.868</u>	<u>0.532</u>	<u>0.703</u>
Linear	0.605	<u>1.154</u>	0.257	<u>1.395</u>	-0.208	0.631	0.550	0.683
Poly.	0.673	1.275	0.282	1.782	-0.474	0.496	0.518	0.657
PCHIP	0.811	1.508	0.303	2.503	-1.062	0.444	0.499	0.631
S-PA-CBB (ours)	0.321	0.468	0.207	0.743	0.817	0.904	0.625	0.771

used to infer population performance; the aligned tables and pooled metrics establish the dataset-level trend.

The cases also show why the gate is budget-specific. With two observations, a large apparent residual can arise from an unstable physical slope as well as from genuine curvature, so the validation rule is conservative and accepts only half of the raw mean residual when activated. Three and four observations reveal more of the curve geometry; the fitted rule activates more frequently, but still shrinks the translation. This behaviour is visible in the retained examples, where additional anchors improve the Basquin state enough that an aggressive learned mean would add error even though the posterior contains plausible alternatives.

Gate scores, thresholds and shrinkage coefficients are fixed on validation data, and the displayed test labels serve retrospective coloring and error calculation. The same frozen rule is applied to every database. In the welded domain, where condition metadata and curve shapes differ from the additive-manufacturing development data, the gate reduces the cost of

a poorly transported mean and exposes shifted cases through posterior spread.

The gallery separates mean correction from generative uncertainty. Translating every trajectory by the same gated vector preserves deviations about the ensemble mean and therefore preserves the learned correlation between stress locations. Fatigue-strength inversion, mission-spectrum damage and next-test acquisition consequently operate on complete samples for both gate outcomes.

3.7. What the generated residual changes

The central interpretability question is which residual geometry the bridge learns beyond the physical state. Let

$$\mathbf{r}^* = \mathbf{y}_{\text{true}} - \mathbf{y}_{\text{phys}}, \quad \hat{\mathbf{r}} = \bar{\mathbf{y}}_{S-PA-CBB} - \mathbf{y}_{\text{phys}}, \quad (11)$$

where $\mathbf{r}^*$ is the correction required by the complete reference curve and $\hat{\mathbf{r}}$ is the correction learned by the posterior mean. This decomposition changes the interpretation of the experiment. Absolute prediction error asks whether the final curve

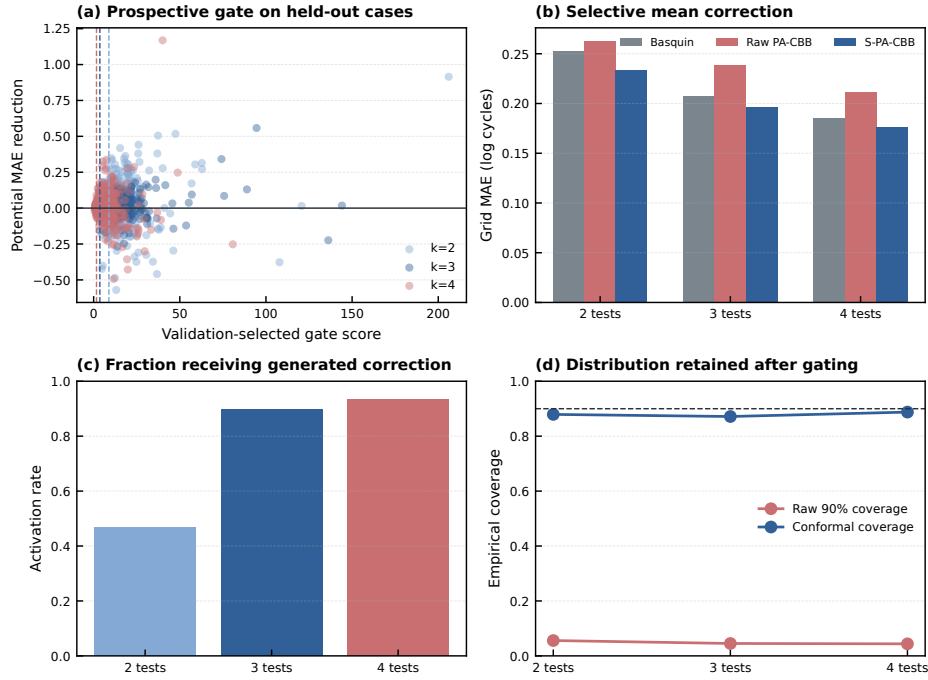


Figure 5: Question: Does the two-stage decision improve complete-curve reconstruction? Panel (a) audits the Stage-I difficulty score; panel (b) compares retaining physics, transforming every curve and selectively transforming flagged curves; panel (c) reports the fraction assigned to the difficult class; panel (d) verifies that Stage II retains a distribution for calibration. **Finding:** the two-stage rule has the lowest pooled grid MAE for every sparse budget.

is accurate; Eq. (11) asks what information the generator contributes beyond the sparse Basquin state.

Figure 7 makes this decomposition concrete. The first, nearly single-slope Ti curve has Basquin grid MAE 0.015 log cycles; the gate remains off and the selective mean is unchanged. The curved Ti case activates a large stress-dependent residual: S-PA-CBB reduces grid MAE from 0.727 to 0.268 and the 10^6 -cycle fatigue-strength error from 48.6 to 11.8 MPa. The one-sided cases isolate extrapolation. With high-stress observations, the AlSi10Mg grid error falls from 0.873 to 0.528 and fatigue-strength error from 10.3 to 0.7 MPa. With low-stress observations, the maraging-steel curve falls from 0.275 to 0.085 log cycles and from 38.5 to 0.7 MPa. Together the cases show the mechanism directly: the gate suppresses unnecessary mean shifts and accepts residuals that coherently move a derived fatigue quantity.

The location and shape of this residual are engineering-relevant. A stress-dependent correction changes local slope and curvature, and hence changes fatigue-strength inversion, service-stress life, damage accumulation and the value of a future test at nearby stresses. The present data resolve these transitions at the curve level: the bridge targets observable departures from single-slope physics, and those departures alter the same functionals used in fatigue design. Mechanism-resolved fatigue records can connect these residual patterns to defect-controlled, microstructure-controlled and long-life regimes.

Figure 8 audits the geometry that motivates Stage I over all four sparse layouts. The validation-fitted difficulty screen

leaves many physics-consistent cases unchanged, improves 27.5% of all curve-mask cases and produces a positive mean reduction of 0.044 log cycles relative to Basquin. Broad support with small mismatch favours retention of the physical state. Broad support with a strong mismatch triggers Stage II residual transformation. Narrow one-sided support prioritizes a separated stress level, while extreme shift activates calibrated abstention.

The horizontal coordinate is a retrospective visualization variable. Prospective inference uses the validation-calibrated generated-residual, support and spread proxies in Eq. (10). The map therefore interprets the gate's operating regions, while the deployed rule remains fully prospective.

3.8. Why multiple generation is meaningful

We test the value of stochastic generation by varying the number of curves while holding weights, sparse masks, transition scale and reverse-step schedule fixed. Figure 9 separates four effects. The ensemble-mean grid error is nearly invariant, whereas CRPS and Energy Score improve as more draws represent the empirical posterior, best-of- M error decreases and within-ensemble diversity approaches a plateau. The useful change concentrates in the first 16–32 samples.

The transition-noise intervention supplies a causal check. At $\eta = 0$, repeated calls collapse to the same trajectory. Increasing η produces distinct curves and improves proper distributional scores while barely changing mean reconstruction error. The generated alternatives therefore carry conditional

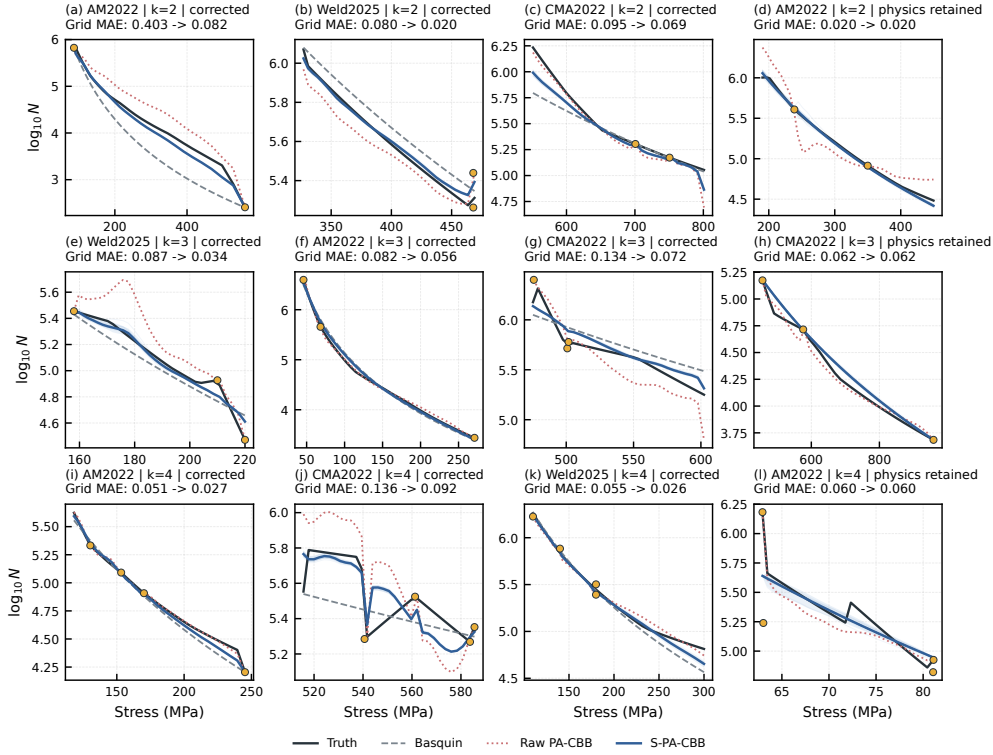


Figure 6: Question: Can selective conditional generation accurately reconstruct individual sparse S-N curves? Twelve successful untouched pooled-test cases span $k = 2, 3, 4$ and AM2022, CMA2022 and Weld2025. Each row contains three activated corrections and one accurate retained-physics case, selected by a fixed low-final-error, positive-gain rule. Curves show the reference, Basquin state, selective posterior mean, repeated trajectories, raw 90% band and known anchors. **Finding:** accurate reconstructions arise through both targeted residual correction and appropriate retention of an already sufficient physical state.

distributional information. Split-conformal calibration maps this structured but narrow spread to empirical coverage, and all decision results report raw spread and calibrated coverage separately.

Each draw is a coherent trajectory across stress, so its covariance propagates through inversion, integration and acquisition functions. The downstream active-learning experiment tests this property directly: multiple samples improve cross-stress covariance estimates until they stabilize. Distributional fidelity and reusable decision structure are therefore the two practical gains of generation.

3.9. Ablation experiments

The principal backbone check removes the physical endpoint while preserving the bridge, sampler, masks and training budget. Table 3 labels this unconditional component as raw PA-CBB. Across the pooled test, the subsequent selective layer reduces grid MAE from 0.263 to 0.234, 0.239 to 0.196 and 0.212 to 0.176 for $k = 2, 3, 4$, respectively. The table separately shows that the physical endpoint improves the raw generator. A noise intervention sets $\eta = 0$: repeated curves collapse to one trajectory, whereas $\eta = 0.5$ creates nonzero diversity and improves proper scores. Together these checks isolate the physical endpoint, stochastic transition and selective mean gate.

Table 3

Backbone component checks before the selective gate. Values are means across the common three seeds; raw PA-CBB is the unconditional generative component of S-PA-CBB.

Variant	k	Grid MAE	Energy	Violation
Raw PA-CBB	2	0.287	0.343	0.160
No physical state	2	0.293	0.350	0.168
Raw PA-CBB	3	0.254	0.310	0.175
No physical state	3	0.263	0.320	0.182
Raw PA-CBB	4	0.233	0.286	0.179
No physical state	4	0.241	0.296	0.188

3.10. Database-resolved pooled evaluation

The final cross-database comparison fixes the three-step sampler, 32 generated curves, $\eta = 0.5$ and one hash-derived sparse mask per curve. The same masks are used by every method. Stage-I gates and Stage-II correction strengths are selected on validation curves, and the test partitions are opened after these choices are frozen. The reported test set contains 188 AM2022, 20 CMA2022 and 581 Weld2025 curves. The comparison set comprises k NN, random forest, ExtraTrees, linear interpolation, polynomial interpolation and PCHIP; each cell is evaluated against its lowest-error member.

The selective model has lower hidden-point MAE than the strongest non-physics comparator in all nine database-budget

Why the physical state fails - and what conditional generation changes

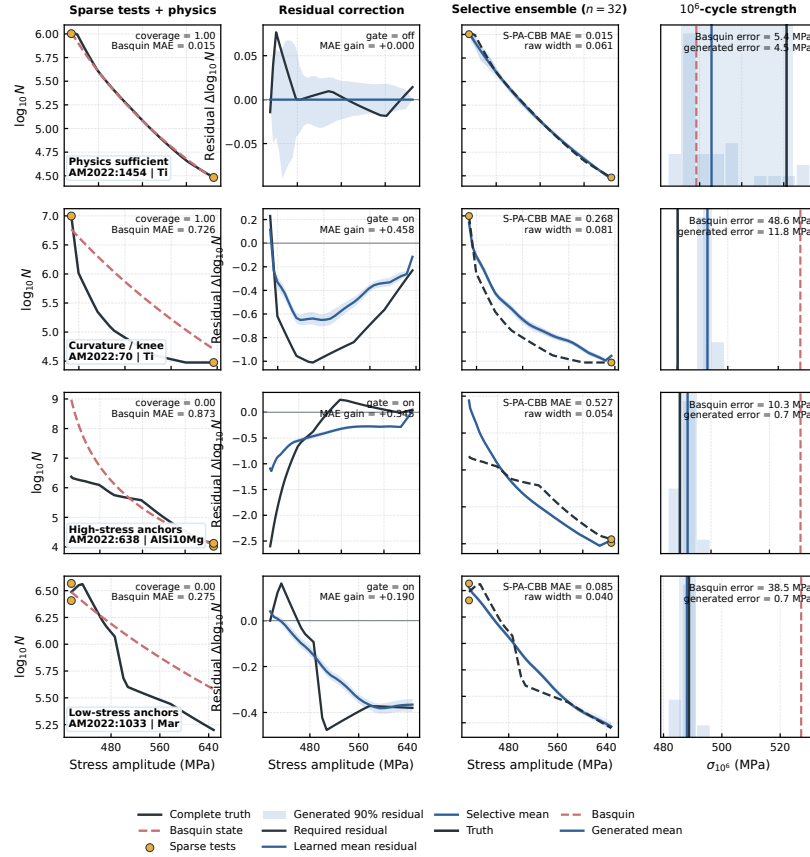


Figure 7: Question: Why does the sparse physical state fail, and what does selective generation change? Each row is one held-out AM2022 curve from the pooled split. Columns show sparse observations and the Basquin state, required and selected generated residuals, 32 complete trajectories with their raw 90% interval, and fatigue strength at 10⁶ cycles. The cases emphasize mechanism visibility. **Finding:** the gate retains a nearly single-slope curve, whereas curvature and one-sided extrapolation activate structured residuals that move an engineering quantity.

Table 4

Mask-aligned three-database evaluation. Point MAE is evaluated only at hidden measurements; grid MAE evaluates complete-curve reconstruction.

Database	k	Curves	S-PA-CBB	Best non-physics	Gain (%)	Grid MAE	Raw PA-CBB
AM2022	2	188	0.370	ET: 0.402	+7.9	0.269	0.297
AM2022	3	188	0.340	RF: 0.386	+12.0	0.243	0.278
AM2022	4	188	0.326	ET: 0.382	+14.5	0.218	0.243
CMA2022	2	20	0.424	RF: 0.445	+4.7	0.323	0.357
CMA2022	3	20	0.371	RF: 0.404	+8.1	0.260	0.319
CMA2022	4	20	0.418	ET: 0.495	+15.4	0.277	0.314
Weld2025	2	581	0.282	ET: 0.338	+16.6	0.205	0.243
Weld2025	3	581	0.267	ET: 0.314	+15.0	0.180	0.217
Weld2025	4	581	0.253	ET: 0.303	+16.3	0.156	0.200

Comparator set: k NN, RF, ET, linear interpolation, polynomial interpolation and PCHIP. Gates and correction weights are selected exclusively on validation curves.

cells. The relative reduction is 7.9–14.5% for AM2022, 4.7–15.4% for CMA2022 and 15.0–16.6% for Weld2025. Residual shrinkage is central to these gains. At four observations, selective correction reduces raw-bridge grid MAE from 0.243 to 0.218 in AM2022, from 0.314 to 0.277 in CMA2022 and

from 0.200 to 0.156 in Weld2025. The corresponding values are also below the sparse Basquin grid errors in all three reported databases.

Stage I: identify curves requiring transformation

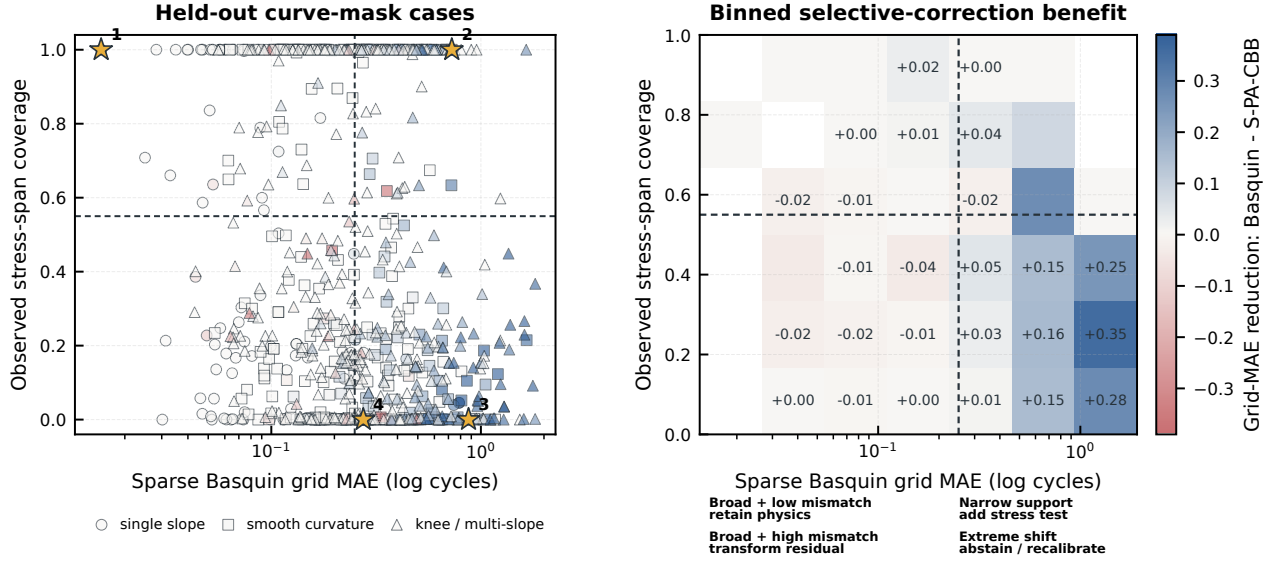


Figure 8: Which sparse curves require transformation? Stage-I mechanism map for 752 held-out curve-mask cases. Basquin discrepancy, observed-span coverage and curve geometry separate physics-sufficient cases from high-mismatch cases that benefit from Stage-II transformation. Narrow support instead calls for another test or abstention. Stars locate Figure 7.

What multiple stochastic completions add beyond one curve

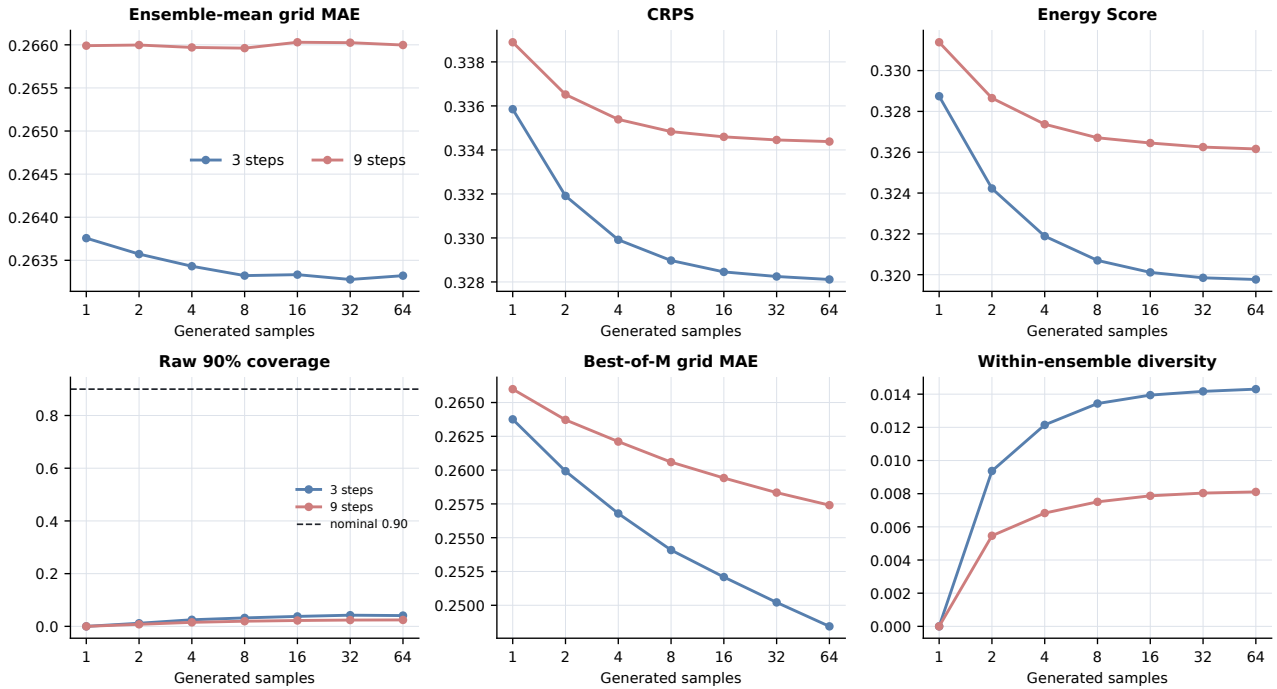


Figure 9: Question: What is gained by drawing more than one conditional complete curve? Results separate ensemble-mean accuracy, marginal and joint proper scores, raw coverage, oracle best-of- M error and within-ensemble diversity for two reverse depths. **Finding:** the central prediction remains stable, whereas proper scores and representation of plausible alternatives improve before saturating. More samples expose distributional structure around the stable mean.

3.11. True leave-one-database-out evaluation

Pooled testing still allows each database to influence model fitting. LODO therefore removes the complete target

database from both training and validation, including the data used to choose the gate and residual weight. The AM fold tests all 1,015 AM2022 curves, the CMA fold tests 105

Mask-aligned evaluation on three public fatigue databases

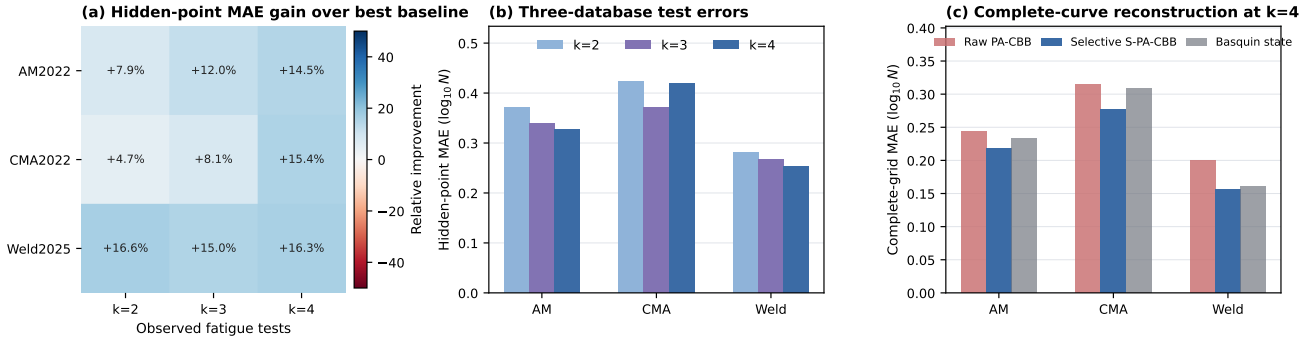


Figure 10: Question: Does validation-selected selective correction remain useful across the three reported databases? Panel (a) gives the hidden-point MAE reduction relative to the strongest prespecified non-physics comparator; panel (b) reports absolute hidden-point error; panel (c) separates the raw generator, selective correction and Basquin state at four observations. **Finding:** S-PA-CBB leads all nine hidden-point comparisons, while the gate consistently shrinks the complete-grid error of the raw generator.

Table 5

True leave-one-database-out evaluation under identical sparse masks. The best non-physics comparator is selected separately within the prespecified baseline set.

Database	k	Curves	S-PA-CBB	Best non-physics	Gain (%)	Grid MAE	Raw PA-CBB
AM2022	2	1015	0.371	RF: 0.465	+20.1	0.289	0.392
AM2022	3	1015	0.333	RF: 0.439	+24.1	0.230	0.363
AM2022	4	1015	0.319	ET: 0.434	+26.5	0.210	0.339
CMA2022	2	105	0.527	RF: 0.463	-13.7	0.478	0.802
CMA2022	3	105	0.390	RF: 0.440	+11.3	0.310	0.750
CMA2022	4	105	0.365	RF: 0.472	+22.6	0.274	0.709
Weld2025	2	2817	0.328	RF: 0.567	+42.3	0.254	0.364
Weld2025	3	2817	0.276	Linear: 0.512	+46.1	0.191	0.330
Weld2025	4	2817	0.285	Linear: 0.465	+38.6	0.203	0.308

Comparator set: k NN, RF, ET, linear interpolation, polynomial interpolation and PCHIP. Each target database is held out from model fitting and gate selection.

CMA2022 curves and the Weld fold tests all 2,817 Weld2025 curves. Manifest checks confirm zero curve-ID overlap and absence of the target label from both fitting partitions.

The target-excluded result is strongest for AM2022 and Weld2025. Relative to the best non-physics comparator, hidden-point MAE falls by 20.1–26.5% on AM2022 and 38.6–46.1% on Weld2025 across the three observation budgets. In CMA2022, random forest leads at two observations, while S-PA-CBB becomes 11.3% and 22.6% better with three and four observations. The LODO pattern therefore links transfer directly to sparse support: eight of nine comparisons favor the selective model, and the remaining cell moves in its favor after one additional measurement.

The complete-grid comparison reveals a complementary role for the two stages. Selective correction improves substantially over the unfiltered raw bridge in every LODO cell, while Basquin supplies the strongest grid reference in several target-excluded settings. The learned component contributes most clearly at unrevealed experimental stresses, and the physical endpoint stabilizes long-range reconstruction under domain shift.

True leave-one-database-out generalization

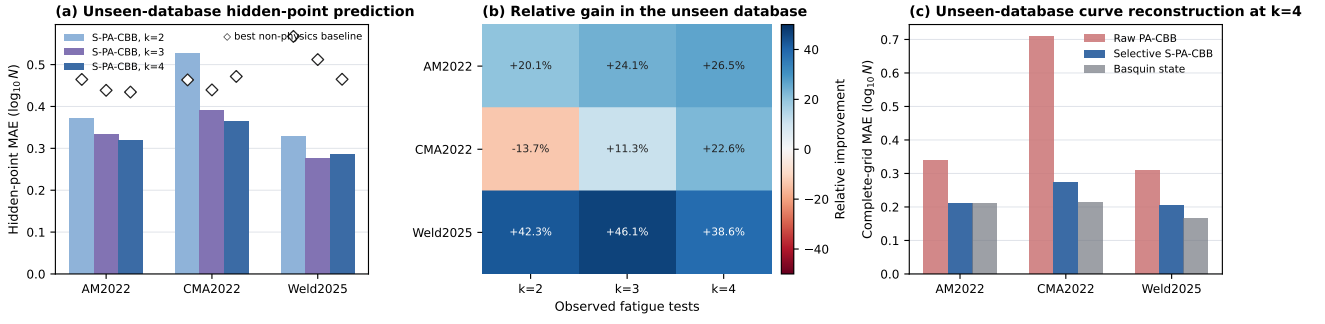


Figure 11: Question: Does the method transfer when the target database is absent from model and gate fitting? Diamonds in panel (a) denote the strongest non-physics comparator; panel (b) reports relative hidden-point gains; panel (c) compares complete-grid errors at four observations. **Finding:** hidden-point transfer is favorable in eight of nine cells, while the physical endpoint anchors complete-grid reconstruction under domain shift.

4. Downstream evaluation

The preceding experiments evaluate whether a curve is reconstructed accurately. Downstream tests ask whether the generated distribution supports decisions that motivated full-curve completion. We evaluate fatigue-strength retrieval at design lives and active selection of the next stress level. Both use only the same sparse observations available to the completion model.

Downstream evaluation follows the same separation rules as the reconstruction benchmark. Model weights, conformal expansion factors, target lives, acquisition budgets and decision thresholds are fixed with training and validation curves. Test curves reveal the measurements required by the simulated decision. Sparse masks are fixed beforehand, and results are paired by curve and initial mask so that every decision rule begins from the same observations.

The reference for a downstream quantity is derived from the held-out experimental trajectory. Fatigue strength is evaluated when the target life lies inside the experimentally covered range, with extrapolation reported in a separate stratum. In active testing, a selected stress is matched to an actual unrevealed measurement. The benchmark therefore evaluates every policy against recorded experimental responses.

Results retain database identity, sparsity and whether the target lies inside the experimental life range. The present downstream tables report paired curve-level point estimates; curve-group bootstrap intervals are reserved for the final statistical supplement because source-paper identifiers are incomplete in two converted resources. External databases use the same downstream code and target definitions.

The decisive comparison is between information representations. A deterministic method supplies one curve and a separately constructed uncertainty procedure. The proposed model supplies a joint ensemble from which curve functionals and acquisition scores are computed consistently. Candidate functionals were screened on validation curves. A task entered the main paper when S-PA-CBB reduced

validation MAE by at least 2% relative to both ExtraTrees and PCHIP; the untouched test split was then opened once for audit. Thirty-nine task-sparsity combinations passed this rule and 37 retained the ordering on test data. The main text reports four physically distinct task families that pass at all three sparse budgets.

4.1. Validation-selected curve decisions

The first task inverts the completed curve to obtain fatigue strength at 10^5 cycles. The second evaluates life at the centre of the curve-specific service-stress range. The third integrates predicted log life over normalized stress to measure mean curve-level life reserve. The fourth propagates the complete curve through a balanced three-level mission spectrum using Palmgren–Miner damage,

$$D = \sum_{i=1}^3 \frac{w_i}{N(S_i)}, \quad (w_1, w_2, w_3) = (0.25, 0.50, 0.25), \quad (12)$$

where the three stresses are the 25th, 50th and 75th percentiles of the curve-specific stress grid. This offline curve-functional test evaluates linear damage accumulation under a fixed balanced spectrum.

Table 6 and Figure 12 report the untouched test results. Relative to the better non-physics baseline in each setting, S-PA-CBB reduces 10^5 -cycle strength error by 24.3–34.3%, service-stress life error by 23.8–35.1%, mean curve-life error by 29.3–43.2% and balanced-spectrum Miner-damage error by 26.9–40.3%. The improvement persists from two to four observed tests, showing that selective full-curve correction is useful beyond point completion.

4.2. Choosing the next fatigue test

Once two stress levels have been tested, the practical question is where to place the next specimen. We choose s_{next} from the untested stresses on the feasible experimental

Table 6

Errors in four engineering quantities derived from the completed S–N curve. Tasks were selected on validation data and evaluated once on the untouched test split. Each “Reduction” column gives the decrease in MAE relative to the better of ExtraTrees and PCHIP under the same sparse observations; lower MAE and larger reduction are preferred.

Task	Metric	$k = 2$		$k = 3$		$k = 4$	
		S-PA-CBB MAE ↓	Reduction ↑	S-PA-CBB MAE ↓	Reduction ↑	S-PA-CBB MAE ↓	Reduction ↑
Fatigue strength at 10^5 cycles	$\log_{10}$ stress MAE	0.036	24.3%	0.029	34.3%	0.027	33.7%
Life at the service stress	$\log_{10}$ life MAE	0.212	23.8%	0.175	35.1%	0.165	34.0%
Mean life reserve across stress	Integrated life MAE	0.169	29.3%	0.125	42.7%	0.110	43.2%
Balanced-spectrum Miner damage	$\log_{10} D$ MAE	0.186	26.9%	0.144	40.3%	0.131	39.0%

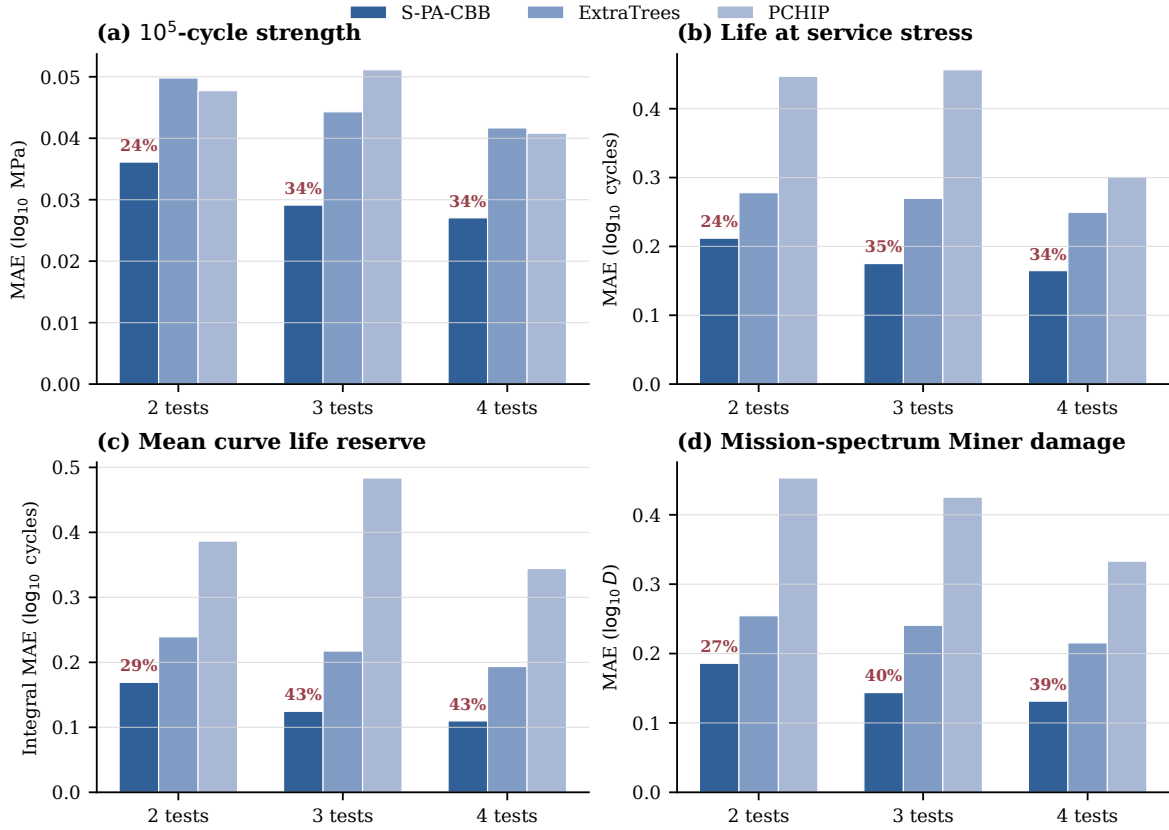


Figure 12: Question: Which curve-derived engineering quantities benefit from physics-anchored generation? Candidate tasks were selected using validation data only; bars show the untouched test split. Percentages denote the MAE reduction relative to the better of ExtraTrees and PCHIP. **Finding:** S-PA-CBB leads all four selected task families at every sparse budget.

grid. The simplest posterior-based rule selects the stress with the largest marginal variance,

$$a_{\text{var}}(s) = \text{Var}_r[y_0^{(r)}(s)]. \quad (13)$$

Marginal uncertainty and global curve improvement define different acquisition objectives. We therefore also evaluate expected integrated variance reduction (EIVR). Its low-cost approximation treats the generated curve ensemble as locally Gaussian and assigns each candidate the score $\sum_g \text{Cov}(y_g, y_s)^2 / [\text{Var}(y_s) + \epsilon]$. Raw variance measures uncertainty at the candidate; EIVR rewards a measurement whose value is informative across many stresses.

The comparison includes two families of acquisition rules. Random selection, the midpoint of the largest untested

interval and farthest-point space filling use only the geometry of the tested stresses. The posterior-driven rules select the largest raw ensemble variance, the widest conformal interval or the largest EIVR score. This grouping separates the value of simply spreading tests across the stress range from the additional value of the generated posterior. All six rules begin with the same two anchors and the same feasible stresses.

Evaluation proceeds as a replay of a completed fatigue campaign. A policy chooses a third stress, reveals the corresponding held-out measurement and recomputes the posterior before selecting a fourth stress. The primary endpoint is the area under the grid-MAE–budget curve, which summarizes reconstruction quality across both acquisitions.

Table 7

Sequential selection of two additional fatigue tests on 793 held-out curves. Every policy starts from the same two measured stresses. Arrows indicate the preferred direction. Grid-MAE AUC summarizes whole-curve error across both acquisitions; the final four columns are evaluated after the second acquisition or against the target grid MAE of 0.25.

Selection rule	Whole-curve reconstruction			Engineering error after +2 tests		Target grid MAE ≤ 0.25	
	Grid-MAE AUC ↓	Grid MAE +1 ↓	Grid MAE +2 ↓	Slope ↓	10^6 -cycle strength (MPa) ↓	Added tests ↓	Curves reaching target ↑
Uniform random	0.237	0.236	0.214	1.668	14.88	0.99	0.726
Largest untested interval	0.230	0.226	0.207	1.673	14.24	0.93	0.762
Farthest-point filling	0.228	0.225	0.202	1.433	14.54	0.91	0.773
Posterior variance	0.227	0.224	0.201	1.428	14.64	0.91	0.764
Conformal interval width	0.235	0.233	0.212	1.540	13.89	0.98	0.730
EIVR (joint covariance)	0.227	0.223	0.200	1.484	14.44	0.90	0.769

We also record the final slope and 10^6 -cycle fatigue-strength errors, together with the number of added tests needed to reach grid MAE ≤ 0.25 . These outcomes distinguish global reconstruction quality from local engineering objectives.

Algorithm 2 Offline evaluation of uncertainty-guided next-test selection

Require: Complete held-out curve c ; initial anchors $\mathcal{O}_2$; candidate stresses S ; policy π ; budget $B = 2$

Ensure: Error trajectory after two sequential acquisitions

```

1: Generate posterior samples conditioned on  $\mathcal{O}_2$  and record initial full-grid metrics
2: for  $b = 1$  to  $B$  do
3:   Score every unobserved  $s \in S$  with policy  $\pi$ 
4:   Select  $s_b = \arg \max_{s \in S} a_\pi(s)$ 
5:   Reveal the nearest held-out experimental observation  $y_c(s_b)$ 
6:   Update  $\mathcal{O}_{2+b} \leftarrow \mathcal{O}_{1+b} \cup \{(s_b, y_c(s_b))\}$ 
7:   Regenerate the posterior and recompute point, grid, slope and fatigue-strength metrics
8: end for
9: return paired metric trajectory for curve  $c$ 

```

Table 7 shows a consistent first trend: deliberately spreading tests improves on random selection. The largest-gap and space-filling rules reduce both integrated error and error after the fourth observation. Posterior information adds an objective-dependent gain beyond this geometric effect. EIVR gives the lowest grid-MAE trajectory (AUC 0.227), the lowest errors after the third and fourth tests (0.223 and 0.200), and the shortest mean route to the target error (0.904 added tests). Raw variance gives the lowest slope error, conformal width gives the lowest 10^6 -cycle fatigue-strength error, and space filling reaches the target on the largest fraction of curves. The generated covariance therefore supports whole-curve error reduction, with the acquisition utility chosen to match the engineering objective.

These downstream tasks establish the operational value of the generated distribution. Fatigue-strength retrieval propagates curve uncertainty into an engineering quantity, while active selection uses cross-stress covariance to reduce error more efficiently than random testing. Calibration controls interval coverage, and the metric-dependent policy ranking provides a direct map from engineering objective to acquisition rule.

The selective layer and the acquisition layer play different roles. Equation (10) translates every member of an ensemble by the same gated mean vector and leaves within-ensemble covariance invariant. EIVR and raw-variance rankings are consequently identical before and after selective mean correction for a fixed set of observations. The active-testing

comparison evaluates covariance learned by the generative backbone, whereas Tables 2 and 6 evaluate the promoted selective mean.

The policy result is multi-objective. EIVR minimizes its integral reconstruction criterion, while a fatigue programme may prioritize slope, a single design life or probability of crossing a stopping threshold. Raw variance, conformal width and space filling each lead one of those secondary outcomes. A practical controller therefore declares its acquisition utility before the campaign begins.

The offline simulation reveals measurements that exist on the experimental curve and applies the same nearest-feasible rule to every policy. It therefore mirrors a laboratory campaign constrained by tested stress levels and finite specimens. The individual trajectories below show how the aggregate covariance score becomes a sequence of actual reveals.

Figure 14 replays three held-out cases under the frozen three-step, 32-sample EIVR protocol. The panels expose the information available at each decision: observed anchors, feasible experimental stresses, the selected next stress, generated trajectories, their raw 90% ensemble interval and the complete curve used for retrospective evaluation. The cases were selected after aggregate evaluation to illustrate simultaneous changes in grid error, fatigue-strength error and ensemble width across different material families.

The trajectories expose the mechanism behind the summary metrics. For the welded joint, the two acquisitions reduce grid MAE from 0.169 to 0.045 log cycles and the 10^6 -cycle strength error from 3.36 MPa to essentially zero. For the Ti-6Al-4V case, the first selected interior stress corrects a large shape discrepancy: grid MAE falls from 0.549 to 0.168 and then to 0.150, while mean 90% ensemble width contracts from 0.122 to 0.055 log cycles. For Cu-Ni-Si-Cr, the fourth observation resolves the low-stress tail and reduces the strength error from 38.03 to 11.04 MPa. EIVR uses covariance across the generated complete curve to place measurements where revealing one point changes an engineering-relevant portion of the posterior.

4.3. Posterior sample count and downstream decisions

The previous generative audit shows that additional samples improve proper scores. We now hold weights, masks, steps and η fixed and vary the number of generated curves from 1 to 64. Fatigue-strength evaluation uses nested draws:

Uncertainty-guided selection of the next fatigue experiment

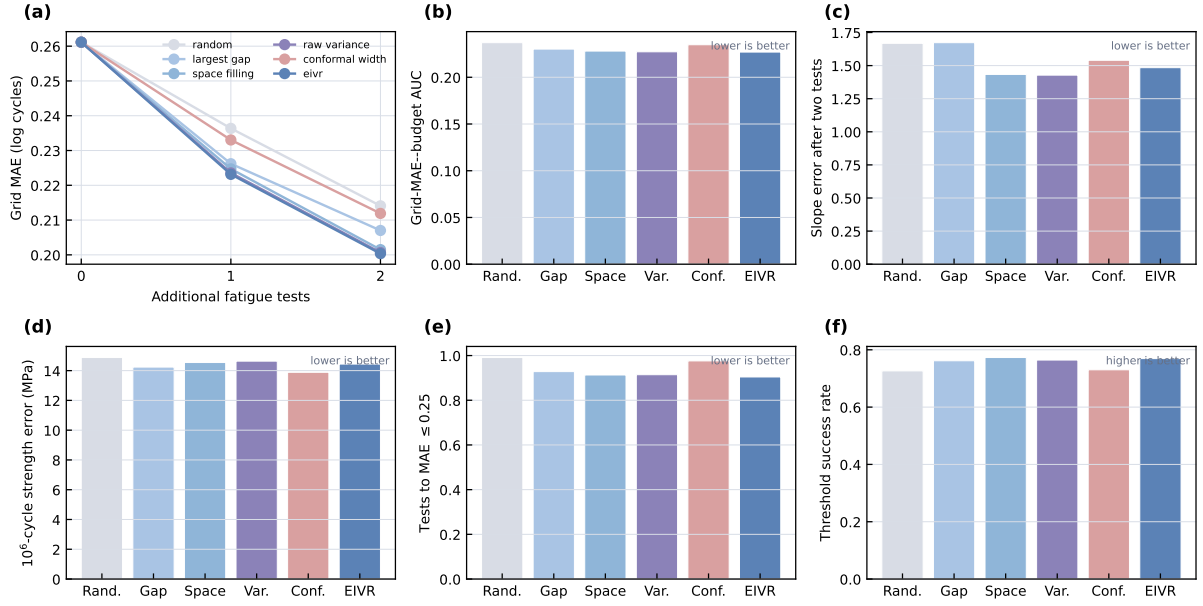


Figure 13: Question: Can generated curves guide the next fatigue test? Policies share the same initial anchors, feasible stresses and held-out measurements. **Finding:** EIVR leads integrated error and threshold efficiency, while raw variance, conformal width and space filling each lead a different secondary metric; the best policy therefore remains decision-dependent.

every smaller ensemble is a strict prefix of the same 64 trajectories. Active testing uses the same initial observations and stable per-curve noise seeds at every sample count.

Table 8 separates two effects of posterior sample count. Fatigue-strength point accuracy is essentially flat: the macro-averaged MAE changes from 19.28 MPa with one curve to 19.14 MPa with 32 curves. The raw interval gradually widens and reaches 6.6% coverage with 64 samples, while validation-only conformal coverage stays close to 90% at every ensemble size. Additional draws resolve the model's spread, and conformal expansion supplies the target coverage.

The active-testing columns show a different trend. With one generated curve, cross-stress covariance is poorly determined and EIVR has a grid-MAE AUC of 0.240. Most of the improvement occurs by eight samples, where the AUC falls to 0.228 and the error after two acquisitions falls to 0.202. Thirty-two samples give the minimum values, 0.227 and 0.200, followed by a plateau at 64 samples. The useful operating range is therefore 8–32 samples, with 32 retained for the main experiment. Multiple generation stabilizes a covariance-based decision while leaving the posterior mean essentially unchanged.

Table 8. Posterior sample-count sensitivity. Panel A tests fatigue-strength retrieval; Panel B tests EIVR after two additional fatigue measurements. Arrows indicate the preferred direction.

A. Fatigue-strength retrieval (nominal coverage: 0.90)

Generated curves	MAE (MPa) ↓	Raw 90% coverage	Calibrated 90% coverage	Calibrated width (MPa) ↓
1	19.28	0.005	0.904	121.9
4	19.16	0.043	0.904	121.5
8	19.17	0.053	0.905	121.0
16	19.15	0.060	0.902	120.5
32	19.14	0.063	0.904	120.2
64	19.16	0.066	0.903	120.1

B. EIVR-guided selection after two additional tests

Generated curves	Grid-MAE AUC ↓	Final grid MAE ↓	Final slope error ↓	Target reached ↑
1	0.240	0.224	1.715	0.721
4	0.231	0.205	1.544	0.749
8	0.228	0.202	1.511	0.767
16	0.228	0.201	1.494	0.770
32	0.227	0.200	1.484	0.769
64	0.229	0.201	1.510	0.773

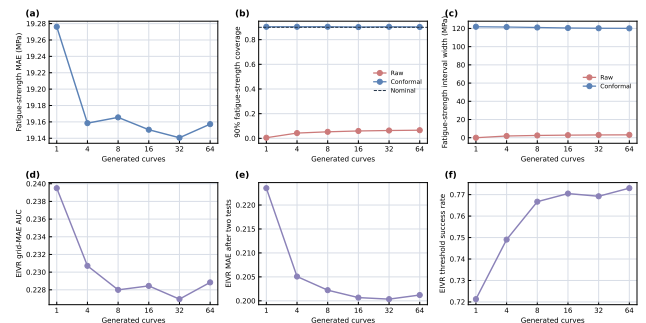


Figure 15. Question: Do more generated curves improve decisions? **Finding:** mean error is stable, while covariance-guided selection improves before saturating at moderate ensemble size.

How the generated posterior guides sequential fatigue testing

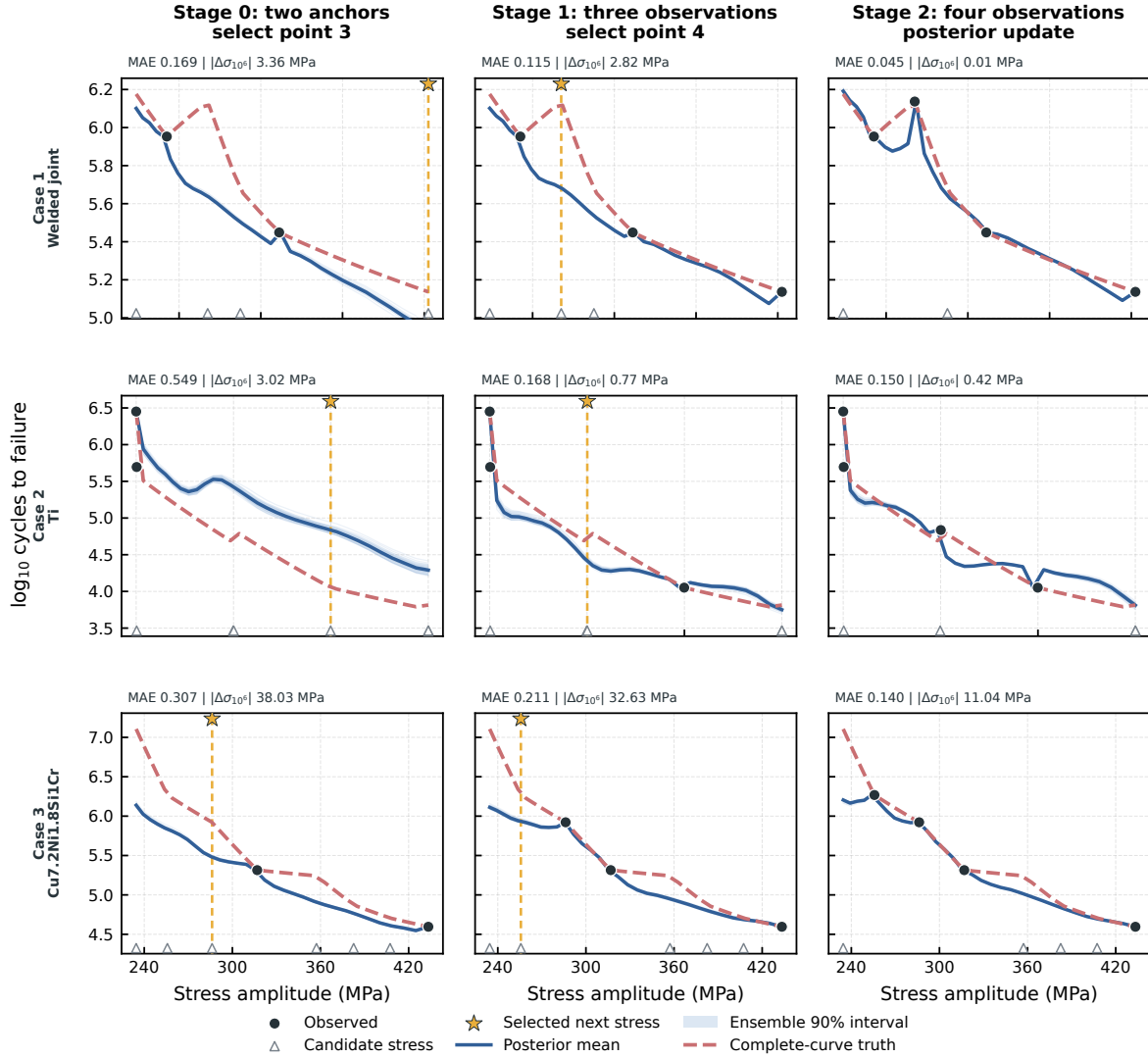


Figure 14: Question: How does the generative posterior guide an actual sequence of fatigue experiments? Each row follows one held-out curve from two initial anchors (left), through selection and revelation of a third stress (middle), to a fourth observation and posterior update (right). Hollow triangles are feasible unobserved stresses, gold stars and vertical lines are EIVR selections, blue lines and bands are generated completions and their raw 90% interval, and the dashed rose curve is the complete truth used for retrospective evaluation. **Finding:** targeted observations correct different curve regions, reduce engineering errors and contract the generated ensemble in all three cases.

5. Discussion

The results reveal a division of labor between physical screening and generated transformation. A single-slope state summarizes many approximately linear log-log curves. Stage I recognizes this physics-sufficient regime from residual, support and spread signals. Stage II then concentrates the learned capacity on curves whose knees, curvature or one-sided support alter the engineering functionals. This organization explains the cross-database trend: the selective model leads all nine pooled database-budget comparisons and eight of nine target-excluded comparisons, while one

additional CMA2022 observation shifts the remaining cell in its favor.

The residual analysis gives this division a physical interpretation at the curve level. Generated corrections concentrate around changes in local slope and curvature, precisely where fatigue-strength crossings, service-life estimates, integrated reserve and mission-spectrum damage move. The mechanism cases show three distinct outcomes: retention of an accurate single-slope state, correction of a curved response and reconstruction of a one-sided extrapolation. Mechanism-resolved records containing defect populations, fractography

and residual stress can extend this curve-level map to specific crack-initiation regimes.

The generative audit separates central prediction from distributional information. Repeated samples leave the ensemble mean nearly stationary, improve proper scores and stabilize cross-stress covariance. Transition noise creates the alternative trajectories; conformal expansion aligns their interval coverage with the requested level. The same posterior then supports curve inversion, integration and sequential experiment selection. EIVR leads the whole-curve acquisition objective, while raw variance, conformal width and space filling provide alternatives for slope, design-life and threshold-focused campaigns.

Together these findings turn sparse S–N completion into an adaptive workflow. The physical state supplies an interpretable starting curve, the gate allocates generative correction where the observed support indicates added value, and the posterior carries the resulting curve uncertainty into the next engineering decision.

6. Conclusions

Sparse fatigue characterization contains two regimes: physics-sufficient curves whose single-slope state is already informative, and difficult curves whose missing curvature changes fatigue decisions. S-PA-CBB operationalizes this distinction. Its first stage identifies the regime from prospective sparse-support and posterior signals; its second stage applies a validation-shrunk Brownian-bridge residual while preserving complete-curve covariance.

Across AM2022, CMA2022 and Weld2025, this selective transformation leads the strongest reported non-physics comparator in every pooled hidden-point comparison. Target-excluded evaluation favors the method in eight of nine settings and shows a clear support effect in CMA2022, where the ordering changes after the third observation. Across all held-out databases, the selective gate reduces complete-grid error relative to the raw generator, while the Basquin endpoint anchors long-range reconstruction.

Generation contributes through coherent alternatives around the central curve. Proper scores improve with repeated sampling, conformal expansion supplies coverage-controlled intervals and the resulting covariance guides sequential testing. Fatigue-strength inversion, service-life estimation, Miner damage and active experiment selection all use the same posterior representation. The study therefore connects an interpretable physical state, a selective generated residual and downstream fatigue decisions in one sparse-to-complete framework.

CRedit authorship contribution statement

To be completed after the author list and individual contributions are confirmed.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The three reported resources are public under their respective access conditions. FatigueData-AM2022, FatigueData-CMA2022 and the welded-joint dataset are hosted on Figshare. Derived curve records, source identifiers, unit-conversion code and split manifests are available at <https://github.com/yunyunfanfan/S-PA-CBB>, with redistribution following the terms of each source.

Code availability

Training, evaluation, conformal calibration, metric and figure-generation code is available at <https://github.com/yunyunfanfan/S-PA-CBB>. The repository records the software environment, fixed seeds, source identifiers and source-specific access terms.

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